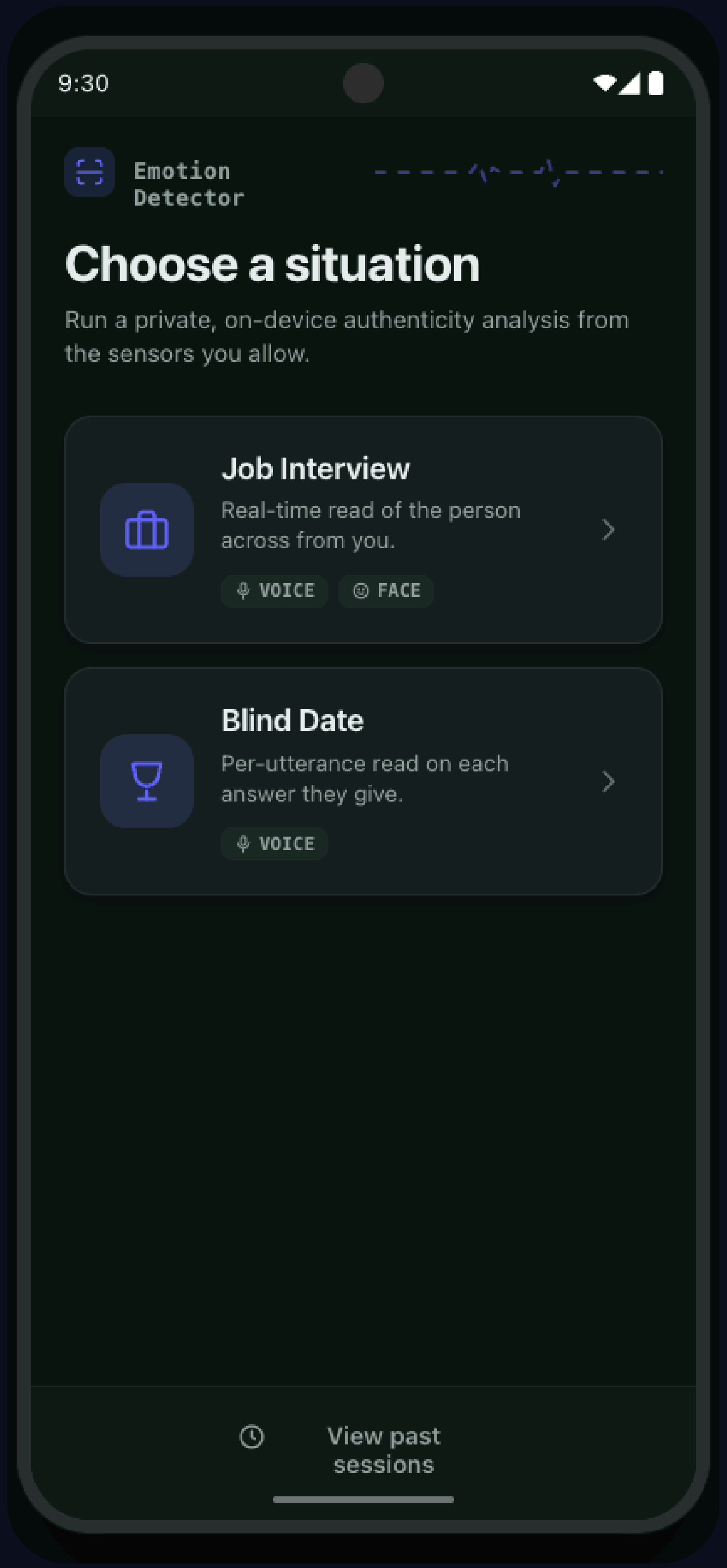
ON-DEVICE · VOICE · FACE · PPG

Emotion and Deception Detector

A private, on-device system that reads **emotion and deception** by fusing voice, facial expression, and heart-rate signals

VOICE FACE PPG DECEPTION



01 —— GOAL & USAGE SCENARIO

Authenticity, read in real time

The goal: an on-device application that acts as a **lie & emotion detector**, surfacing whether what a person says matches what their body reveals.

Job Interview

Capture the person's face.
Voice and facial expression are read continuously as they answer.

VOICE FACE

Blind Date

The other person rests a fingertip over the rear camera and flash. We pair that fingertip pulse with their voice for a continuous genuine / fake read.

VOICE PPG

Two pillars: per-signal models, a sensing app

The Model

Per-signal models, **combined per scenario** rather than one end-to-end fused net.

How it is trained — per-modality models on labelled affect data.
How it is combined — voice+face for Interview, voice+fingertip-PPG for Blind Date.

The Application

An Android app that **senses** live signals and runs inference fully on-device.

What it senses — microphone, camera, and fingertip PPG heart-rate.
How it serves — sliding-window inference, no network round-trip.

Trained per signal, read per scenario

Job Interview

NO PPG

VOICE Log-mel → CNN	→	EMOTION HEAD 8-class
VOICE Log-mel → CNN	→	VERACITY HEAD genuine / fake
FACE 96×96 crop → CNN	→	EMOTION HEAD 8-class

App compares **face vs voice emotion** side by side + a **Face–Voice mismatch** score; the voice model drives the deception gauge.

Blind Date

+ FINGERTIP PPG

VOICE + FINGERTIP PPG Log-mel + pulse → CNN / 1-D	→	EMOTION HEAD 8-class
VOICE + FINGERTIP PPG Log-mel + pulse → CNN / 1-D	→	VERACITY HEAD genuine / fake

App outputs a **continuous genuine / fake read** plus the dominant emotion.

Voice, face, and a fingertip pulse

Voice

Microphone captures prosody & tremor
Take the most recent 5-second window (2.5 stride)
Convert it into a log-mel spectogram
Same transform used in training
(n_fft 2048, hop 1024, per-window z score)

Face

Camera detects the face and crops it (96×96) to read the surface emotion being shown.

Used ML Kit face detection to find the face box

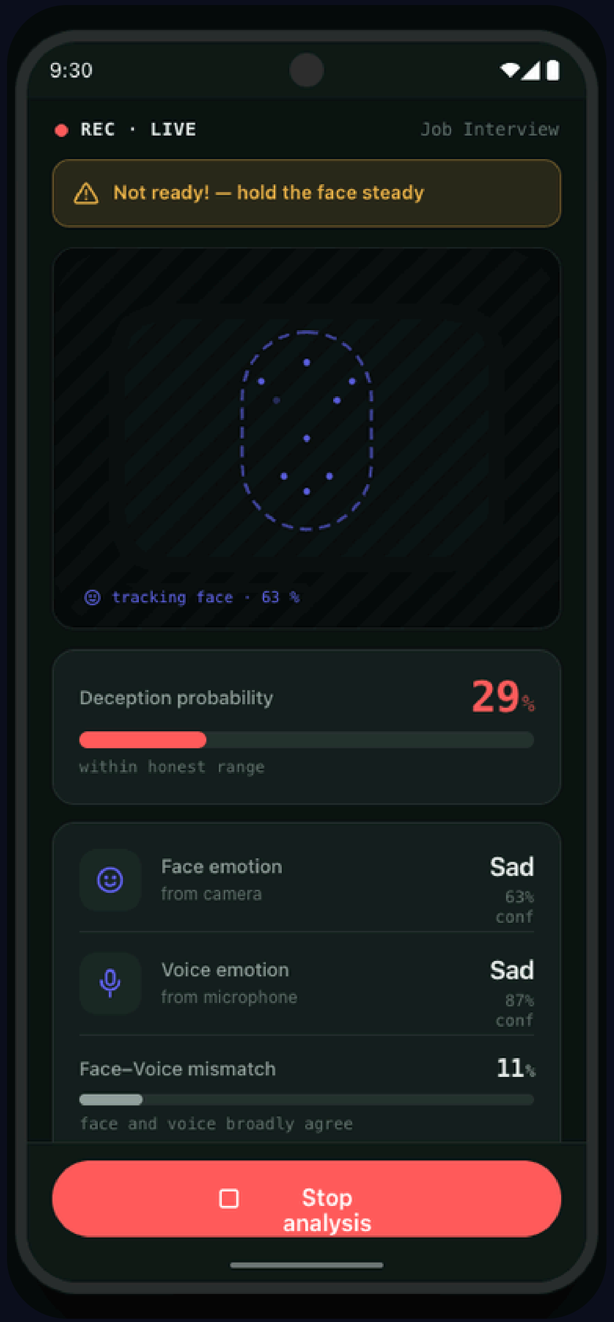
Fingertip PPG

Heart rate from a fingertip resting on the rear camera and flash — a red-channel pulse extracted from the live frames.

Privacy by design. All three streams are processed and discarded on the device — nothing is uploaded or stored off-phone.

05 —— APP FUNCTIONALITY

Two inference modes, one live read-out



VOICE + PPG

Blind Date mode

Lightweight: a fingertip pulse lock plus the mic, giving a continuous genuine / fake read and emotion.

CAMERA + VOICE

Interview mode

Face emotion and voice emotion combine into a live deception gauge plus a face–voice mismatch indicator.

LIVE GAUGE

Deception %

TELL-TALE

Face–Voice mismatch

SURFACE

Emotion + conf.

Sliding window & fingertip PPG lock

Sliding-window inference

Continuous signals are sliced into overlapping windows, so a verdict updates every step instead of waiting for the clip to end.

Fingertip PPG lock policy

A policy gates when the pulse is sampled — only while the fingertip covers the lens and the red channel is stable.

- **RUN** Finger covers the lens, red channel stable → sample PPG, refresh BPM.
- **HOLD** Finger lifts or signal noisy → reuse last stable BPM.
- **WARN** No pulse lock → prompt the user to place a fingertip.

07 —— BEFORE / AFTER OPTIMIZATION

Shrinking the model to fit the phone

Pruning

Remove low-importance weights / channels to cut compute with minimal accuracy loss.

Quantization

Shipped as float16 TFLite (FP32 → FP16, 2× smaller); swap to INT8 figures only if measured.

METRIC	BEFORE	AFTER	Δ
Model size	XX MB	XX MB	↓ −%
Latency / window	XX ms	XX ms	↓ −%
Peak memory	XX MB	XX MB	↓ −%
Accuracy	XX.X%	XX.X%	−0.X pt

Drop in your measured numbers — the layout scales to whatever you report.

08 —— DEMO

See it run

Timeline & Contributions

PHASE 1 (5.11 - 5.24)
Data & model design

PHASE 2 (5.24~5.31)
Training (per-modality models)

PHASE 3 (5.24~5.31)
App & sensing

PHASE 4
Optimize & demo

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App design & engineering — UI, sensing pipeline, on-device inference, data preprocessing (mel-spectrogram, ML kit face detection)

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Model Plan & training — per modality, accuracy analysis
Optimization - pruning & quantization.

We have collected for 35 participants!
(from class, outdoor, friends, etc)

Thank You !

VOICE FACE PPG DECEPTION ON-DEVICE

박준병 & 최재웅 · MCA Final Presentation